
Dynamic k-Space-Aware Coil Compression for Accelerated MRI

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Abstract

Modern magnetic resonance imaging (MRI) systems utilize large receive coil arrays to enhance signal-to-noise ratio (SNR) and enable parallel imaging, but this results in massive raw data sizes that burden storage and I/O bandwidth. Standard coil compression techniques, such as PCA-based decoupling, effectively reduce channel counts but are often inefficient because they treat all frequency components equally. This project introduces a dynamic, k-space-aware coil compression framework that adapts PCA rank and bit allocation to local frequency content. By implementing PCA-based coil decoupling followed by dynamic k-space masking and waterfilling bit allocation, the proposed method optimizes the rate-distortion tradeoff. Experimental results demonstrate that this approach yields a superior Pareto frontier compared to JPEG, DCT, and uniform PCA baselines, preserving sharper edges and higher structural similarity (SSIM) at lower bit rates.

1 Introduction

Modern magnetic resonance imaging (MRI) scanners routinely employ multi-channel receiver arrays with 32–64 coils to improve signal-to-noise ratio (SNR) and to enable parallel imaging. While these additional coils significantly benefit image quality and acquisition speed, they also inflate the size of the raw data, creating bottlenecks for archiving, transmission, and large-scale processing pipelines.

A common remedy is *coil compression*, which maps original coil channels into a smaller set of virtual coils. Standard approaches, particularly PCA-based array compression, are effective at reducing channel counts but suffer from a critical limitation: they are *frequency-agnostic*. By applying a single global transformation matrix across the entire k-space, these methods treat sparse, high-frequency peripheral data with the same importance as the dense, energy-rich low-frequency center. This rigidity leads to suboptimal compression efficiency.

To break this limitation, we propose a *Dynamic k-Space-Aware Coil Compression* framework. Unlike uniform PCA, this method adapts the compression rank and bit allocation according to local frequency content. By decoupling the coils and applying dynamic masking, we allocate more degrees of freedom to the conditioning-sensitive center frequencies while aggressively compressing the redundant periphery.

Our experimental results demonstrate that this proposed method yields a superior Pareto frontier compared to standard baselines, including JPEG, DCT-based compression, and PCA-based uniform compression. These findings validate the hypothesis that k-space energy is non-uniformly distributed across coil channels, and that exploiting this spectral unevenness allows for sharper edge preservation and higher structural similarity (SSIM) at lower bit rates.

2 Literature Review

This project draws on concepts from parallel MRI, k-space sampling, coil compression, and classical transform coding.

MRI Background and Forward Model: The acquisition process in parallel MRI is modeled as a linear system where the underlying object $x \in \mathbb{C}^{H \times W}$ is encoded by multiple receiver coils. The forward model generates multi-coil k-space data $y_c \in \mathbb{C}^{H \times W}$ via coil sensitivity maps $S_c \in \mathbb{C}^{H \times W}$ and a binary k-space subsampling mask $P_\phi \in \{0, 1\}^{H \times W}$. This relationship is expressed as:

$$y_c = P_\phi \mathcal{F}(S_c \cdot x), \quad (1)$$

where $\mathcal{F}$ denotes the discrete Fourier transform and c indexes the coil channel ($c = 1, \dots, C$). The mask P_ϕ typically follows a variable-density Poisson-disc sampling pattern [1, 2, 3], a standard compressed sensing approach. This acquisition model is illustrated in Figure 1.

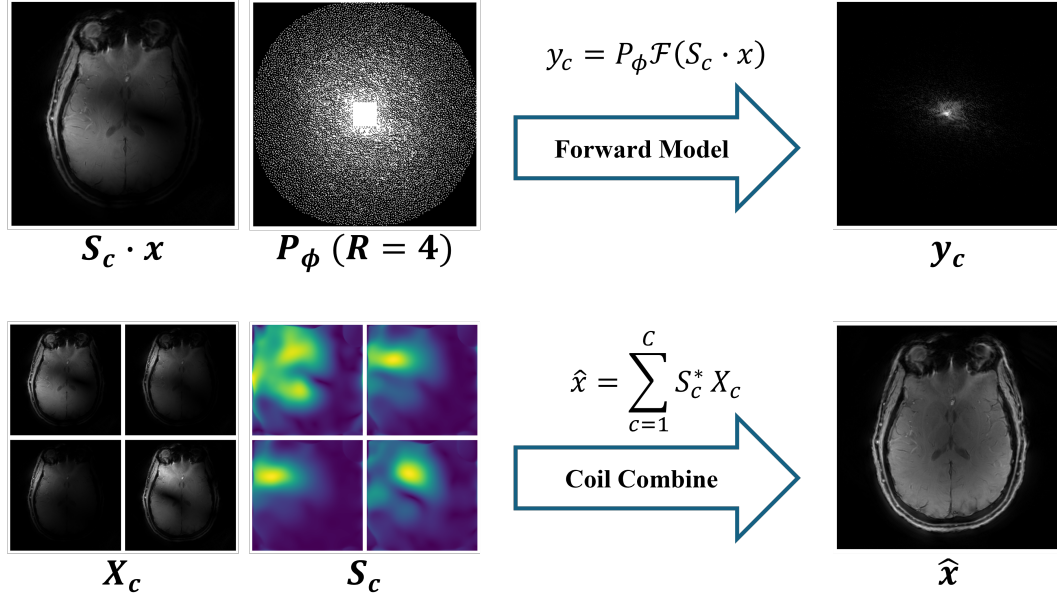


Figure 1: **MRI Forward Model and Coil Combination.** The forward model generates multi-coil k-space data y_c from the underlying object x via sensitivity maps S_c and a subsampling mask P_ϕ . The reconstruction process involves combining these coil images to form the final complex image $\hat{x}$.

Coil Combination and Reconstruction: To recover the image from y_c , we utilize sensitivity maps S_c , which are estimated from a central calibration region using ESPIRiT [4]. Denoting the fully sampled complex multi-coil image by $X_c = \mathcal{F}^{-1}(y_c)$, the final combined complex image $\hat{x}$ is formed by a standard sensitivity-weighted sum:

$$\hat{x} = \sum_{c=1}^C S_c^* X_c. \quad (2)$$

The goal of this project is to compress the raw data y_c such that the quality of this final reconstructed image $\hat{x}$ is maximized for a given bit budget.

Coil Compression and Transform Coding: To evaluate our proposed method, we compare against standard approaches in coil compression and transform coding. PCA-based coil compression [5] reduces channel count by diagonalizing the coil covariance matrix. Classical image compression schemes such as JPEG [6] use blockwise 2D DCT.

Within this project, these ideas are instantiated as three reference compression baselines:

- **JPEG on real/imag mosaics:** Real and imaginary parts of each coil are tiled into mosaics and compressed with standard JPEG across multiple quality factors.

- **DCT compression with water-filling:** A global 2D DCT is applied to real/imag parts of all coils, and the largest coefficients by magnitude (across all coils) are kept and entropy-coded.
- **PCA-based uniform coil compression:** A global coil covariance is estimated, and coil vectors are projected onto the top- K principal components; PCA coefficients and basis vectors are quantized and compressed.

3 Methods

The proposed *Dynamic k-Space-Aware Coil Compression* framework operates directly on the acquired multi-coil k-space data y_c . The compression pipeline implements a non-reversible lossy encoding strategy to reduce redundancy and remove irrelevant high-frequency noise, followed by a standard lossless encoding stage (zlib) to minimize the final bitstream size.

3.1 Algorithm Pipeline

The method is implemented as an encoder-decoder pair. An overview of the three-stage pipeline is shown in Figure 2.

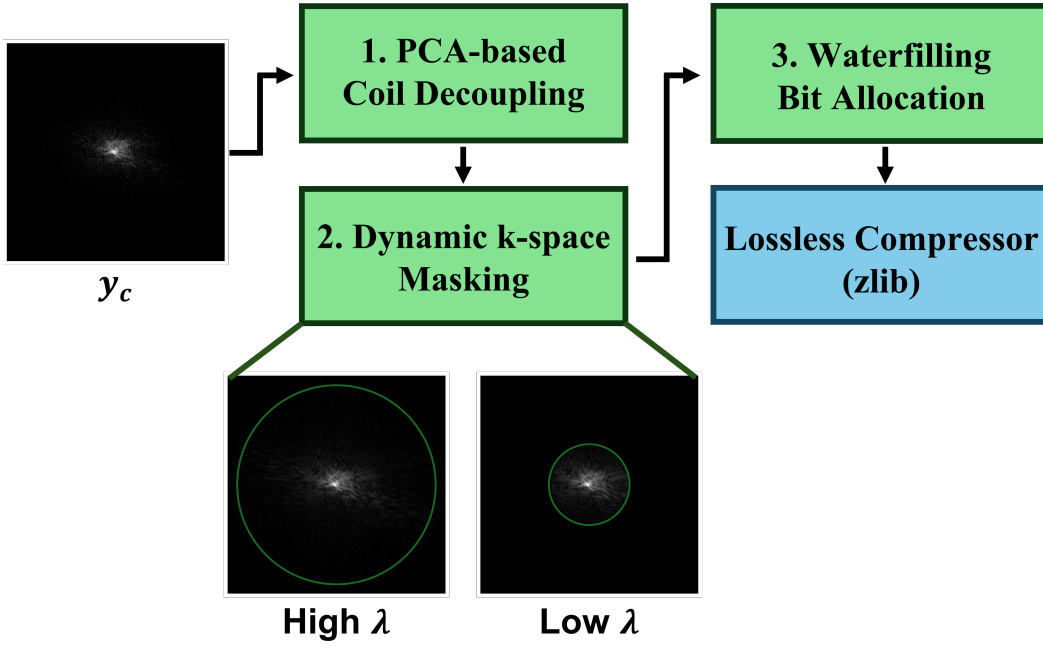


Figure 2: **Overview of the proposed method.** The pipeline includes PCA-based coil decoupling to create virtual channels, followed by dynamic k-space masking that varies with channel importance, and finally a waterfilling-based bit allocation strategy.

1. PCA-based Coil Decoupling: We first decorrelate physical coils into virtual eigen-channels. Given the input k-space data y_c , we estimate the global coil covariance and compute its eigendecomposition. This maps the C correlated physical coils into C orthogonal virtual coils ordered by singular value magnitudes λ_k .

2. Dynamic k-Space Masking: We define concentric radial bands in k-space $\mathcal{B}_b$. Unlike uniform compression, we apply variable-radius circular masks to the virtual coils based on their importance (λ_k):

- **Dominant Coils (High λ):** Retain full k-space details.
- **Minor Coils (Low λ):** Retain only low-frequency contrast ($k \in \mathcal{B}_{low}$), masking out high frequencies to zero.

3. Waterfilling Bit Allocation and Quantization: To optimize quantization for the remaining unmasked coefficients, we employ a waterfilling algorithm. The optimal bit rate R_k for the k -th virtual channel is derived from rate-distortion theory:

$$R_k = \max \left(0, \frac{1}{2} \log_2 \frac{\lambda_k}{\theta} \right) \quad (3)$$

where θ is a threshold parameter found via binary search for a target total bitrate. A scaling factor $S_k \propto 2^{R_k} / \sqrt{\lambda_k}$ is applied to the virtual coil data to normalize the variance relative to the quantizer step size. The scaled data is uniformly quantized to the nearest integer. Finally, the resulting stream of integers is encoded using the **zlib** lossless compression library to produce the output bitstream b .

3.2 Implementation and Code Quality

The entire pipeline is implemented in Python. The codebase is maintained in a public GitHub repository: <https://github.com/wogur110/EE274-dynamic-coil-compression>.

- **Documentation & Reproducibility:** The repository contains a detailed README.md explaining environment setup, usage instructions, and steps to reproduce the experiments. Inline comments are used extensively throughout the code.
- **Unit Testing:** Modular unit tests were implemented for each compression method (JPEG, DCT, Uniform PCA, and the proposed dynamic pipeline) to verify the functional correctness of the individual encoder-decoder pairs.
- **Tools:** AI-assisted coding support was provided by Cursor (Gemini 3 Pro) for boilerplate generation and debugging.

4 Results+Analysis

4.1 Experimental Setup

We evaluated the methods on fully sampled complex 64-coil brain MRI data (294×294). The forward model included k-space subsampling with acceleration factors ranging from $R = 1$ to $R = 16$. For completeness, Rate-Distortion (R-D) curves for all acceleration factors ($R \in \{1, 2, 4, 8, 16\}$) are provided in the Appendix. Performance was measured using SSIM and PSNR on the final ESPIRiT-combined complex images.

4.2 Quantitative Analysis (Rate-Distortion)

We generated R-D curves plotting PSNR (left) and SSIM (right) against bitrate to benchmark performance.

Importance of Coil-Aware Compression: Figure 3 displays the results for acceleration factor $R = 2$, comparing six distinct methods: JPEG (blue), DCT (orange), Uniform PCA (red), Uniform PCA+Waterfilling (green), Dynamic Masking (purple), and Dynamic Masking+Waterfilling (brown).

A key observation is that all coil-compression variants consistently outperform the non-coil-aware baselines (blue and orange curves), confirming that explicitly modeling inter-coil correlations is essential. Furthermore, the R-D curves are notably non-monotonic and jagged. This behavior arises because the reconstruction metric (ESPIRiT combination) is highly non-linear; small changes in quantization levels for specific eigen-channels can interact unpredictably with the sensitivity map combination, causing fluctuations in the final image quality metrics.

Impact of Dynamic Masking and Waterfilling: Figure 4 illustrates the results at a high acceleration factor of $R = 16$. The PSNR curve (left) exhibits unusual behavior, failing to increase monotonically with bitrate. This anomaly stems from the extreme sparsity of the k-space data at $R = 16$, which makes signal recovery ill-conditioned. However, the SSIM curve (right) follows the expected trend seen at Figure 3, providing a more reliable measure of structural fidelity.

The ablation study in Figure 4 reveals distinct roles for each component. **Dynamic coil compression (purple)** proves most effective in the *low bit-rate regime*, where aggressively masking redundant peripheral noise in minor coils preserves essential structure. Conversely, **Waterfilling (green)**

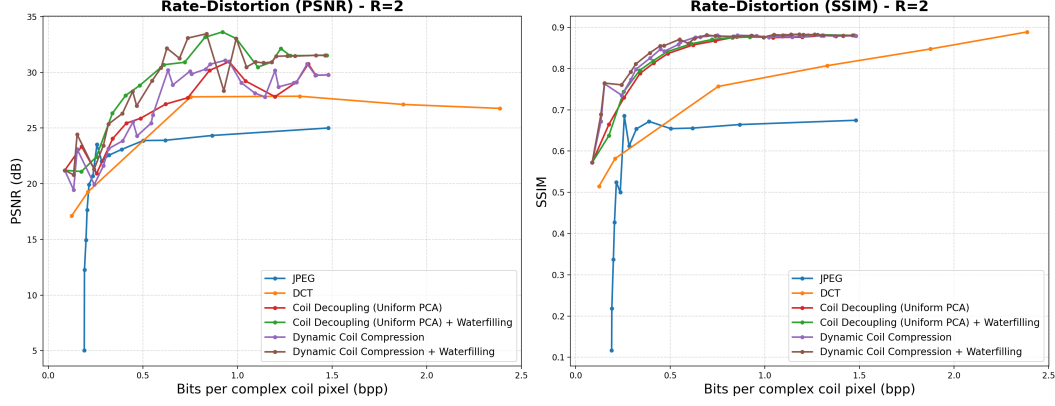


Figure 3: **Rate-Distortion Comparison at $R = 2$.** Comparison of PSNR (left) and SSIM (right) curves. All four coil-compression based methods (red, green, purple, brown) significantly outperform the image-domain JPEG (blue) and transform-domain DCT (orange) baselines. This gap underscores that explicitly modeling inter-coil correlations is the dominant factor for efficient multi-coil MRI compression.

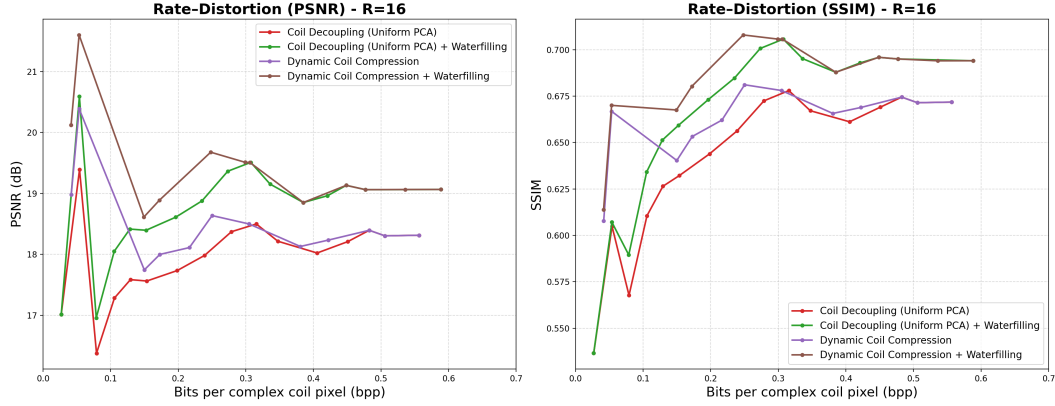


Figure 4: **Dynamic k-space Masking and Waterfilling at $R = 16$.** Rate-Distortion curves comparing standard Coil Decoupling against the proposed method. The proposed method (brown) yields the superior Pareto frontier compared to Dynamic Masking alone (purple) or Uniform PCA (red).

becomes critical in the *higher bit-rate regime*, where it efficiently distributes the generous bit budget to the most significant eigen-channels. Combining both techniques (the proposed **brown curve**) synergistically leverages these strengths, making the overall compression significantly more effective across the entire operational range.

4.3 Qualitative Analysis

Visual comparisons at acceleration factors $R = 1$ and $R = 2$ are presented in Figure 5.

- **Artifacts:** JPEG and standard Coil Decoupling baselines exhibit blurring and aliasing artifacts at low bit rates.
- **Structure Preservation:** The proposed method preserves sharper edges and finer structural details. The dynamic masking ensures that even for minor coils (which contain little energy but some structural info), the low-frequency contrast is preserved, as evidenced by the lower intensity in the error maps.

Qualitative Comparison of Compression Methods ($R = 1$ and $R = 2$)

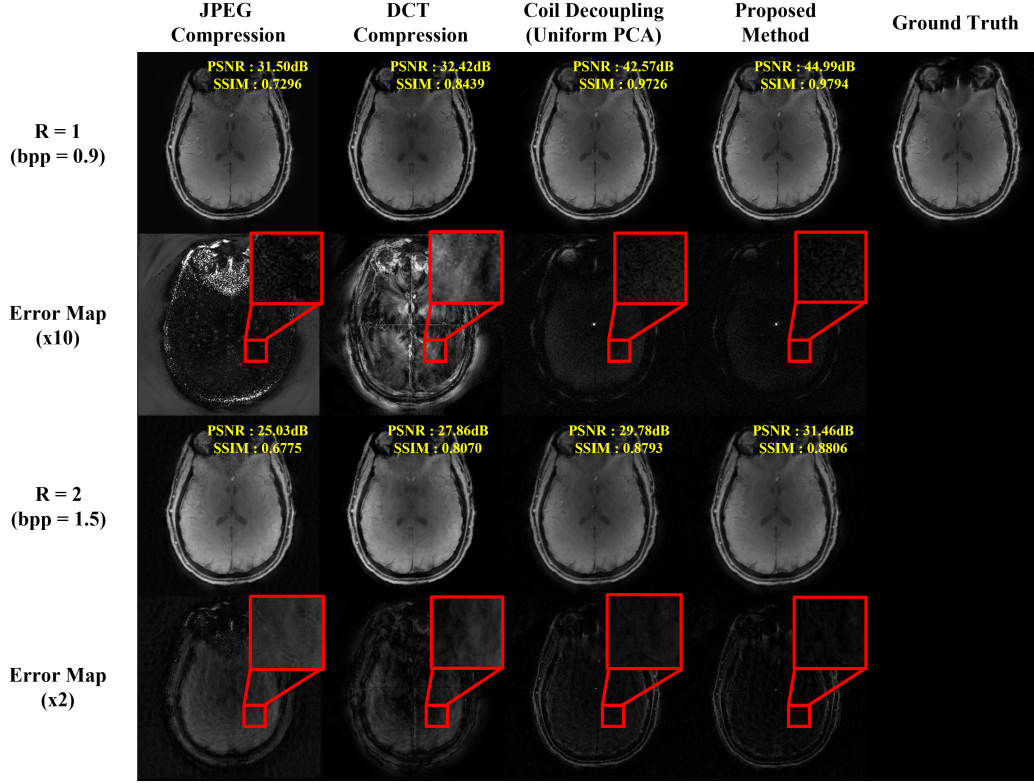


Figure 5: **Qualitative Comparison of Compression Methods** ($R = 1$ and $R = 2$). The figure displays reconstructed images, corresponding error maps, and zoomed-in regions for JPEG compression, DCT compression, Uniform PCA, and the proposed method. The proposed method significantly reduces aliasing artifacts and preserves sharper edge details compared to the baselines.

5 Conclusions

In this work, we introduced a *Dynamic k-Space-Aware Coil Compression* framework designed to overcome the limitations of frequency-agnostic compression in multi-coil MRI. By moving beyond the rigid global transformation of uniform PCA, our method adapts to the local spectral content of the data. We achieved this by integrating three key components: PCA-based coil decoupling to isolate virtual channels, dynamic masking to aggressively compress high-frequency redundancy in minor coils, and a waterfilling algorithm to optimally allocate bits based on signal variance. This approach effectively prioritizes the conditioning-sensitive low-frequency information that is crucial for accurate sensitivity-weighted reconstruction.

Our experimental results validate that this frequency-adaptive strategy yields a superior Pareto frontier compared to standard baselines. Quantitative analysis shows that the proposed method consistently achieves higher SSIM for a given bit budget, particularly in the challenging low-bitrate regimes where JPEG and standard coil decoupling fail to preserve structural fidelity. Qualitative comparisons further confirm that our approach significantly reduces aliasing artifacts and maintains sharper edge definitions, even at high acceleration factors ($R = 16$). These findings underscore the importance of explicitly modeling both inter-coil correlations and k-space energy distributions in MRI data compression.

Future work will focus on implementing this coil compression scheme within advanced reconstruction frameworks, such as compressed sensing and deep learning-based reconstruction (e.g., VarNet), to further reduce scan times while maintaining diagnostic quality.

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References

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A Appendix: Extended Results

This appendix provides the full set of Rate-Distortion (R-D) curves for all acceleration factors tested ($R = 1, 4, 8, 16$), supplementing the $R = 2$ results presented in the main text.

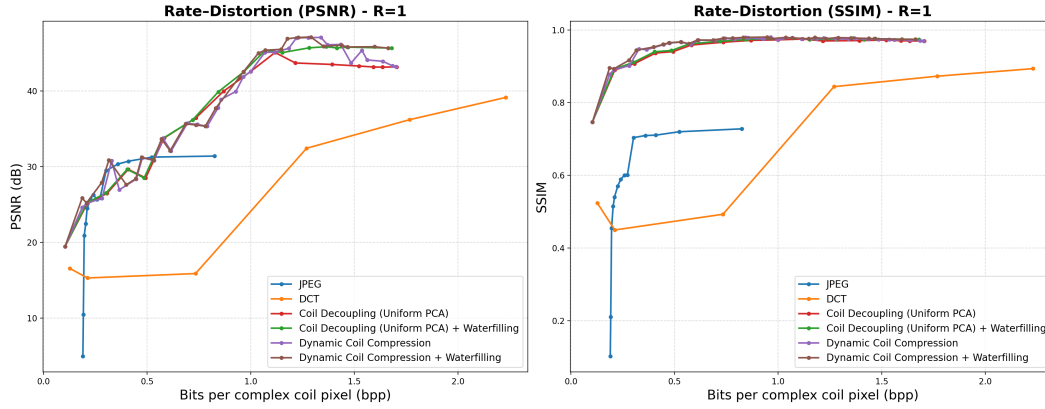


Figure 6: **Rate-Distortion Comparison at $R = 1$ (Fully Sampled)**. Comparison of PSNR (left) and SSIM (right) curves.

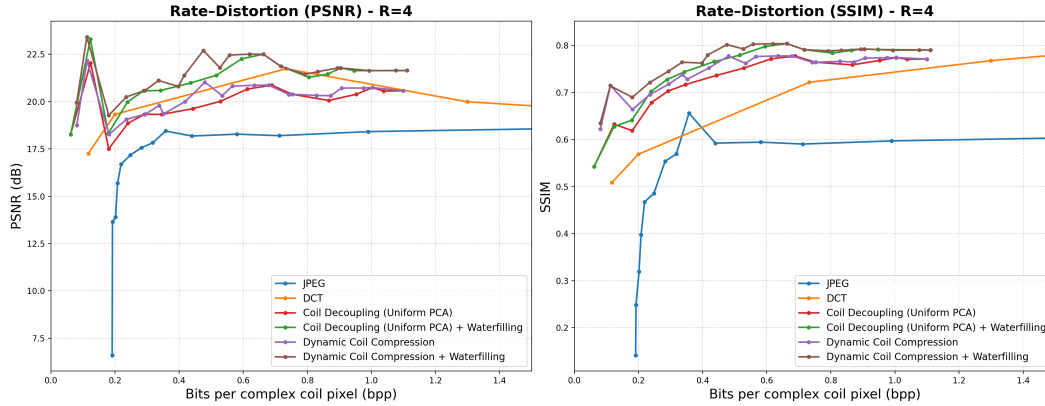


Figure 7: **Rate-Distortion Comparison at $R = 4$** . Comparison of PSNR (left) and SSIM (right) curves.

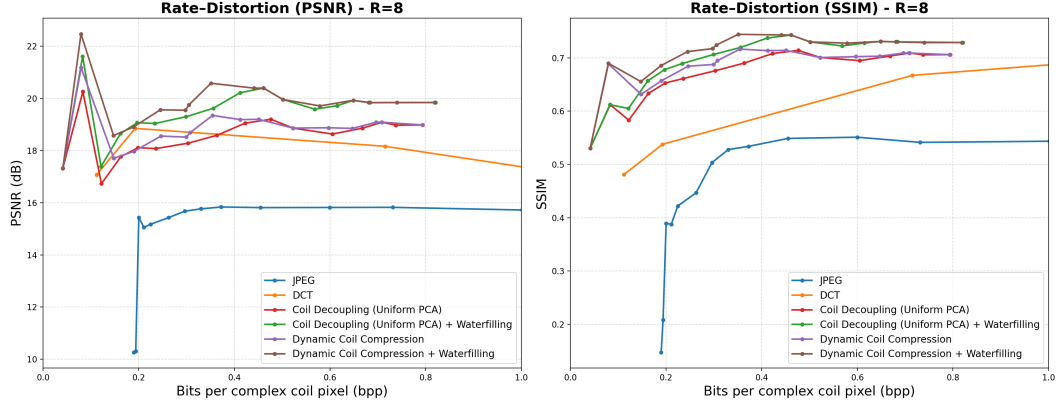


Figure 8: **Rate-Distortion Comparison at $R = 8$.** Comparison of PSNR (left) and SSIM (right) curves.

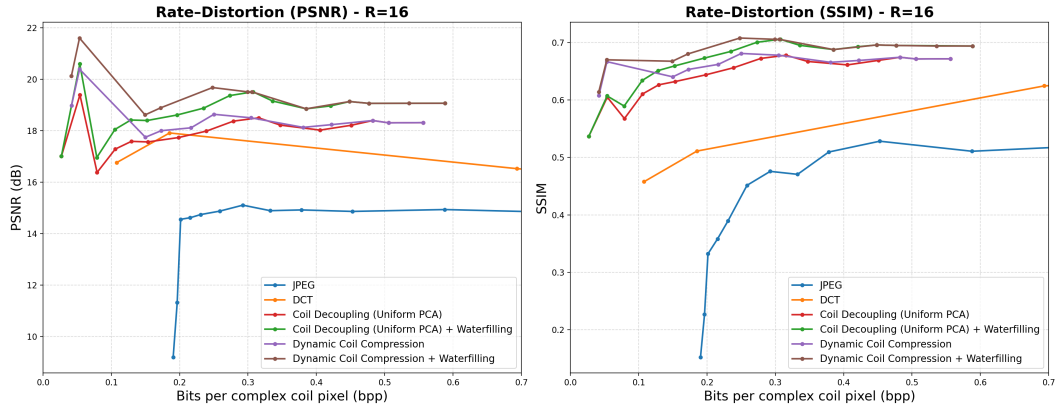


Figure 9: **Rate-Distortion Comparison at $R = 16$.** Comparison of PSNR (left) and SSIM (right) curves.